PALEOMAGNETISM OF THE BOOT-PHANTOM PLUTON AND THE AMALGAMATION OF THE JUVENILE DOMAINS IN THE PALEOPROTEROZOIC TRANS-HUDSON OROGEN, CANADA

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The 1838 Ma Boot-Phantom pluton outcrops over ~18 km^2 in the Amisk collage of the Flin Flon belt in the Paleoproterozoic Trans-Hudson orogen (THO) of northern Saskatchewan. It contains several petrologic phases ranging from gabbro to tonalite. It is massive and unmetamorphosed, the youngest major pluton in the collage, and it was emplaced at 1838 Ma into low-grade metamorphic rocks. Paleomagnetic analyses using alternating field and thermal step demagnetization methods were run on 229 specimens from 24 sites in and adjacent to the pluton. Its characteristic remanent magnetization direction (ChRM) is shown from thermal demagnetization and saturation isothermal remanence analyses to reside mostly in single domain to multidomain magnetite, commonly with high unblocking temperatures near the 585°C Curie temperature, but sometimes in pyrrhotite. The ChRM of the pluton is shown to be primary from contact tests and is directed at Decl. = 46.7°, Inc. = 83.2° (α_{95} = 4.1°, k = 72, N = 18 sites) and includes both normal and reversed polarities. Vertical late-phase diabase and felsic dike sites intersect about at right angles, indicating that the pluton has not been tilted since emplacement. The Boot-Phantom pluton's paleomagnetic pole is at 84.6°W, 62.4°N (dp = 7.9°, dm = 8.0°). This pole gives a polar paleolatitude for the Amisk collage, consistent with all other ~1850 Ma poles from terranes on the southeast side of the Manikewan suture to the Superior craton. It provides, along with the Reynard Lake pluton's pole, an apparent polar wander path segment for the Amisk collage. This path is compared with coeval segments from the Superior craton, Hanson Lake block and Lynn Lake-La Ronge domain. Together the segments display the classical paleomagnetic signature for a collisional orogen with the crustal allochthons of the southeastern THO accreting to the Superior craton at ~1830 Ma.

1. Introduction

During the past two decades it has been realized that the Trans-Hudson Orogen (THO) is composed of a number of domains that were assembled during the Paleoproterozoic as the Archean Superior and Hearne cratons collided to close the intervening Manikewan ocean (Fig. 1) (e.g. Lewry 1981; Stauffer 1984; Hoffman 1990). Based on scant paleomagnetic data, Dunsmore and Symons (1990) suggested that this ocean was ~5000 km wide at

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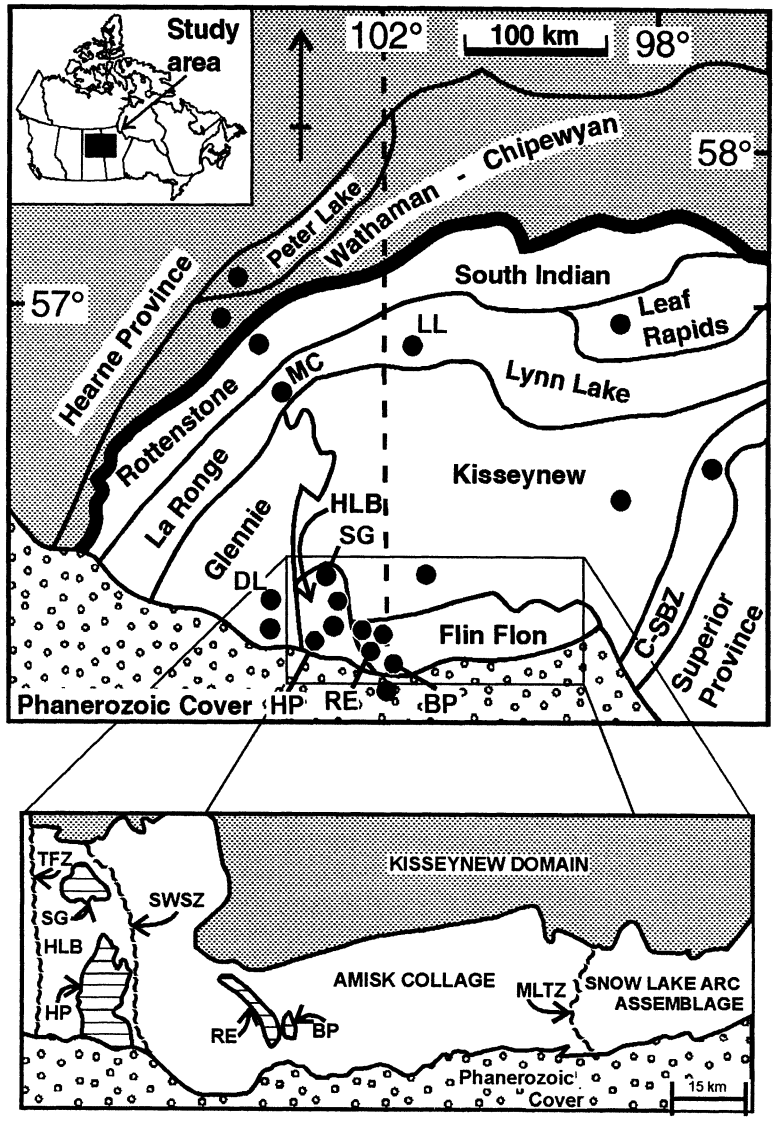


Figure 1. Map of the Trans-Hudson orogen after Hoffman(1988) and Lucas et al (1996). Bold abbreviations: C-SBZ, Churchill-Superior boundary zone; HLB, Hanson Lake block; MLTZ, Morton Lake thrust zone; SWSZ, Sturgeon-weir shear zone; and, TFZ, Tabbernor fault zone. Paleomagnetic collections shown as solid symbols. Those with measured pole positions used in this paper are: (collection unit, accronym, longitude, latitude, one sigma deviation, number of sites averaged, (reference)): Boot-Phantom pluton, BP, 85°W, 62°N, 5°, 18 (this study); Deschambault Lake pegmatites, DP, 84°W, 68°N, 5°, 17 (preliminary values; D.T.A. Symons, pers. comm); Hanson Lake diorite pluton, HP, 95°W, 36°N, 6°, 16 (Gala et al, 1994); Lynn Lake gabbro pipes, LL, 85°W, 9°N, 4°, 29 (Dunsmore and Symons 1990); Macoun Lake pluton, MC, 85°W, 56°N, 10°, 12 (south lobe; Symons et al 1994); Molson dikes, C component, Mo_c, 141°W, 27°N, 3°, 19 (Zhai and Halls 1994); Reynard Lake pluton, RE, 107°W, 40°N, 6°, 18 (Symons 1995); Sahli felsic granulite, SG, 84°, 49°N, 5°, 19 (Gala et al 1997); and, Subbury irruptive complex SC, 115°W, 42°N, 3°, 17 (Morris 1984). The heavy line on the southeast side of the Wathaman-Chipewyan batholith marks the Manikewan suture.